

Agenda — 08-02-2025

FPN → Feature Pyramid Network →



features

→ 0.1
→ 0.2
→ 0.3
→ 0.7



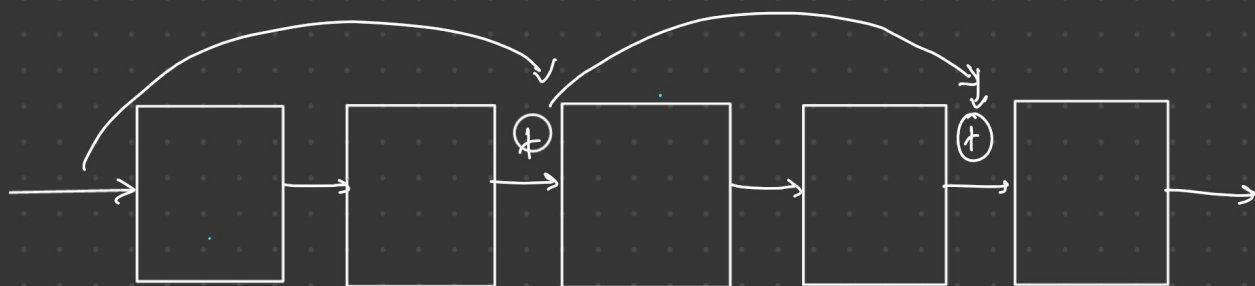
person

not person

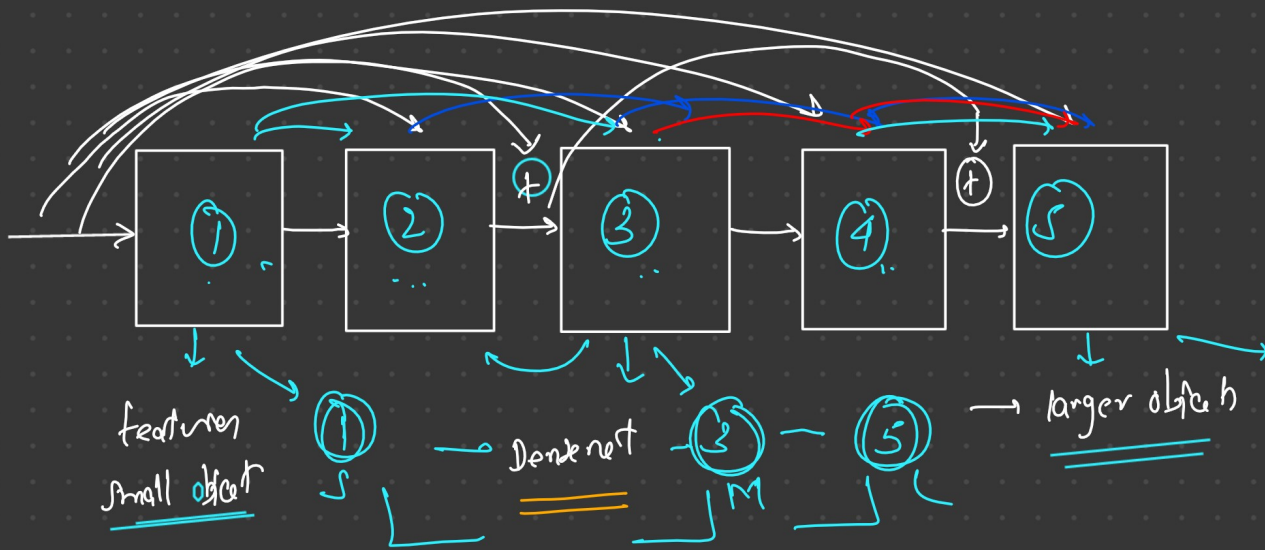
, not cat

→ smaller details / objects

— maxpooling →
— compression →
— high resolution feature



Resnet



Feature Pyramid Network:

Multi-scale feature representation technique used in OD models, yolo v3, v4, faster-RCNN.

→ Improve detection especially for small objects. → combining feature from multiple levels of CNN.

Step 1: pick a Backbone. → feature Extraction.

Resnet 50 - Resnet 18, Resnet 101 FPN

Darknet 53 - Yolo

CSP Darknet 53 - Yolo

v6 v6-16

Step 2:

BB → feature Map → FPN

Layer → $\left(\frac{I - f + 2p}{s} \right) + 1$ ← Feature Map size

Input

Conv, 2, 3x3, p1 (c1) → $\frac{512 - 3 + 2(1)}{2} + 1$ ↙ edges, texture

Conv, 2, 3x3, p1 (c2) →

Conv, 2, 3x3, p1 (c3) → small object

Conv, 2, 3x3, p1 (c4) → larger object

Conv, 2, 3x3, p1 (c5) →

HD
↓
small
objects

256 x 256

128 x 128

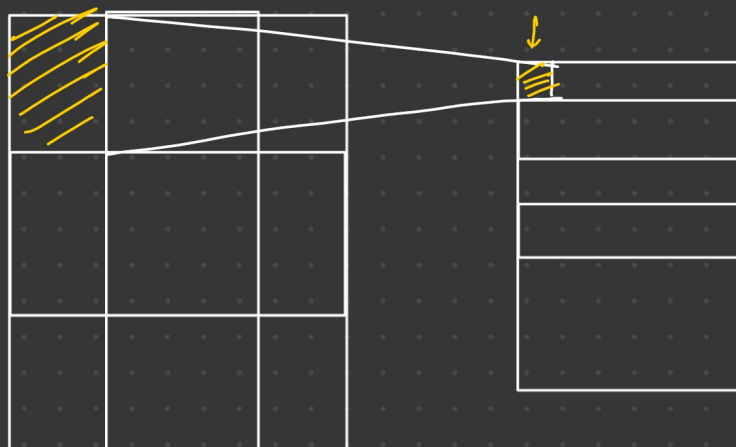
64 x 64

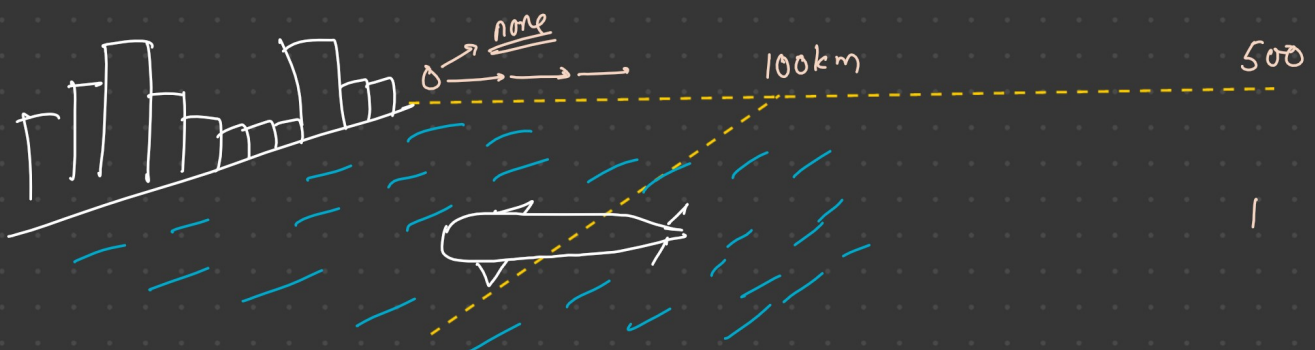
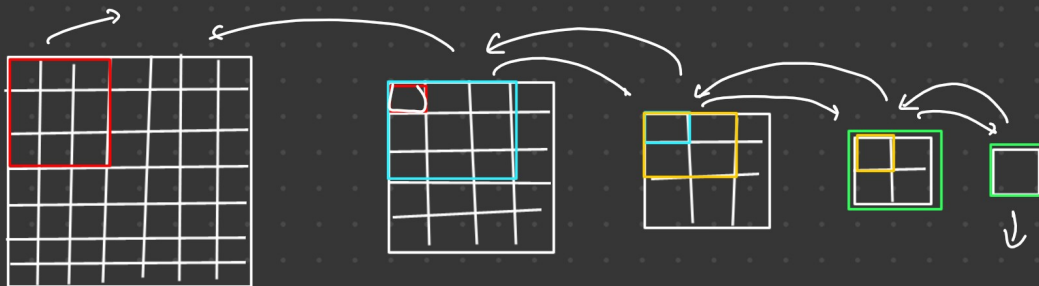
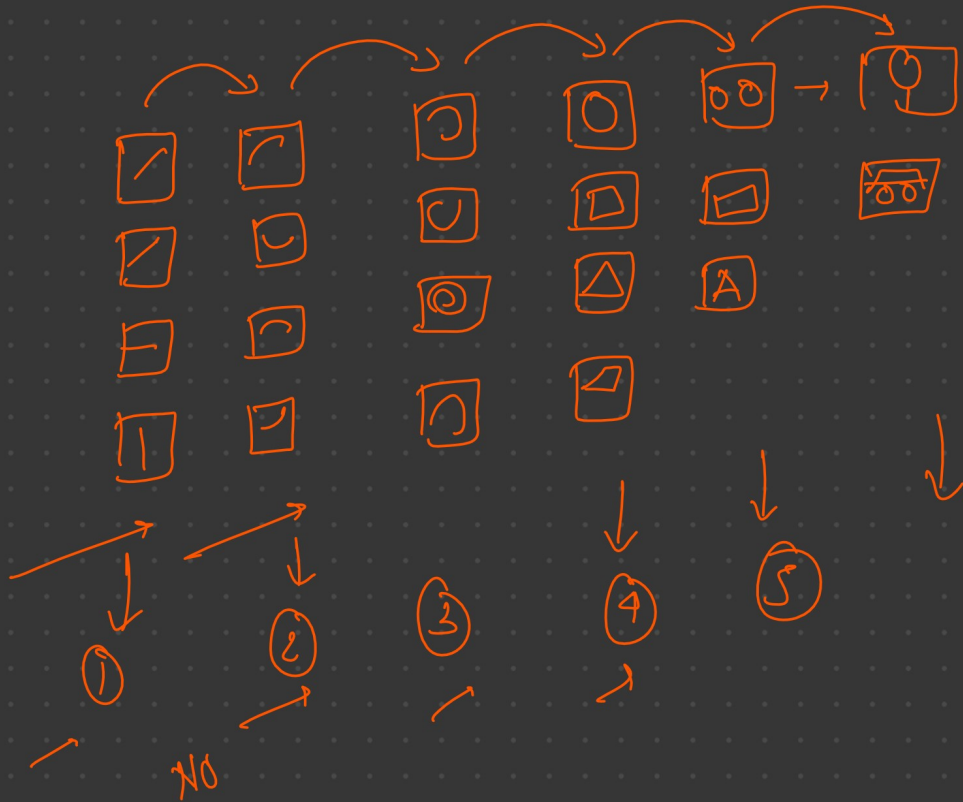
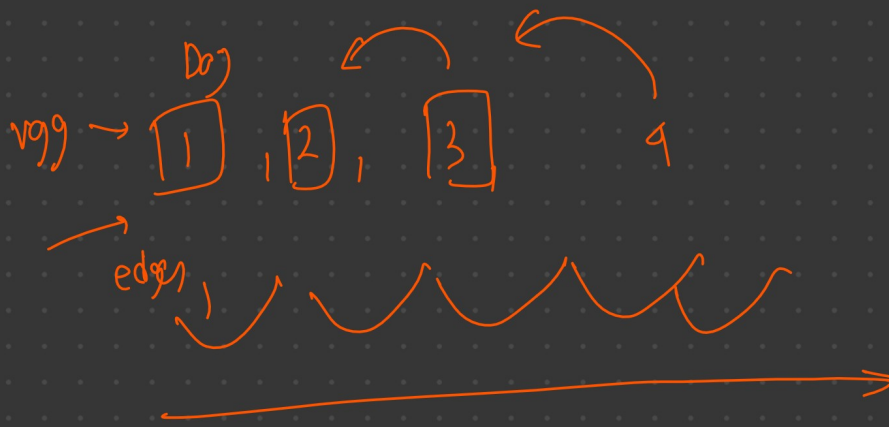
32 x 32

LD

↓
large
objects

16 x 16





Receptive field grows because each convolution sees more pixels of the original image as we deeper.

$c_1, c_2 \rightarrow$ Small receptive field

$c_3, c_4, c_5 \rightarrow$ Large receptive field.

Layer	Feature Map size	
Input	512×512	1×1
Conv, 2, 3×3 , P1 (c_1)	256×256	$3 \times 3 \rightarrow$ <u>object</u>
Conv, 2, 3×3 , P1 (c_2)	128×128	7×7
Conv, 2, 3×3 , P1 (c_3)	64×64	$15 \times 15 \rightarrow$
Conv, 2, 3×3 , P1 (c_4)	32×32	$31 \times 31 \rightarrow$
Conv, 2, 3×3 , P1 (c_5)	16×16	63×63

$$Rf_L = Rf_{(L-1)} + (k-1) \times \left(\sum_L s \right)$$

$$c_1 = (1 + (3-1)) \times \textcircled{1} = 3 \swarrow$$

$$c_2 = (\textcircled{3} + (3-1)) \times \textcircled{2} = 3 + \underline{2 \times 2} = 7$$

$$c_3 = 7 + (2) \times \textcircled{4} = 15$$

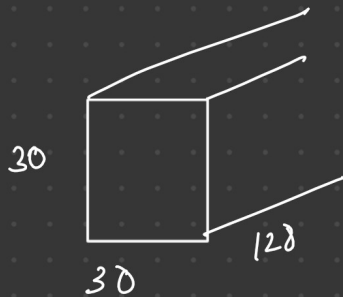
$$c_4 = 15 + (2) \times 8 = 31$$

$$c_5 = 31 + (2) \times 16 = 63$$

C3, C4, C5
 ↑ ↑ ↑



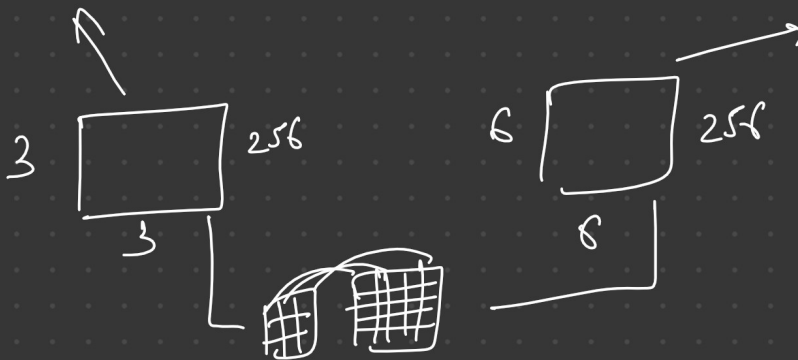
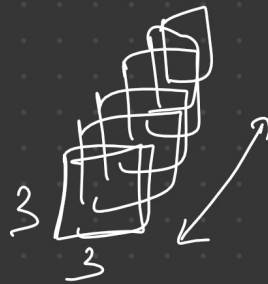
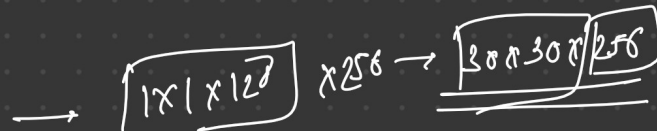
$$P_L = \underline{1 \times 1 \text{ conv}(C_L)} + \boxed{\text{Upsample}(P_{L+1})} \rightarrow \underline{\underline{256}}$$



30x30x1

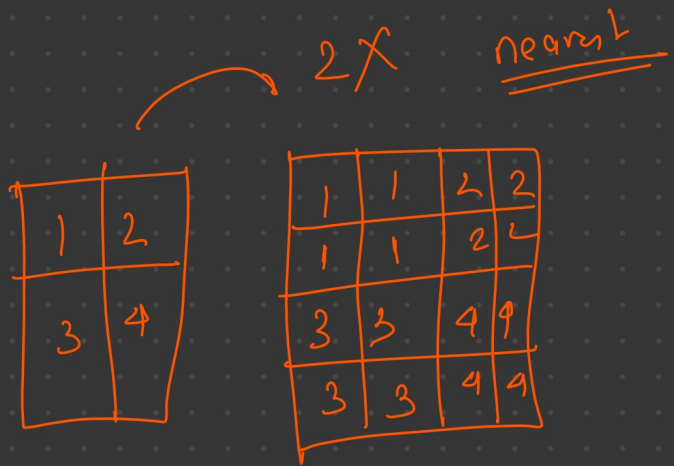
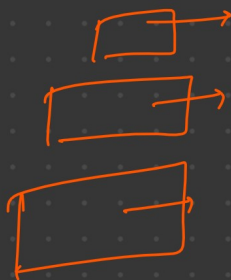
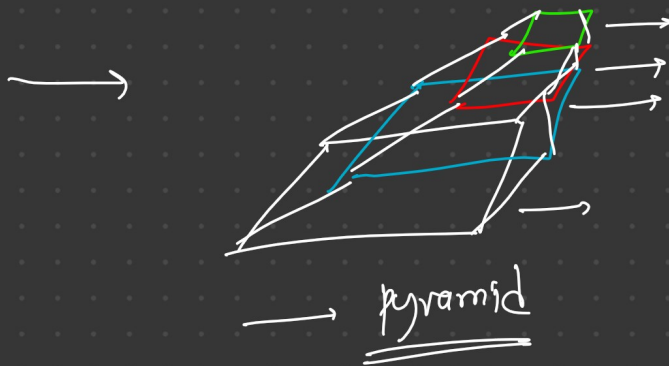
$1 \times 1 \rightarrow$ decrease feature channel

$\rightarrow$ increase feature channel.



3x3 3x3
 6x6 8x8

= P5 input C5 16x16 No upsample output 16x16 → large object XL
 = P4 C4 32x32 Yes P4 = P4 + upsample P5 → large object L
 = P3 C3 64x64 Yes P3 = P3 + upsample P4 → medium
 = P2 C2 128x128 Yes P2 = P2 + upsample P3 → small object



FPN

